Reading List — σ-Essential Problem

Patrick S. Mahon
Personal reference

June 2026

The object: a concrete σ-complete non-Boolean irreducible OML carrying a finite two-valued state on a $\perp$-closed subposet that extends to no global σ-additive two-valued state. Localized to σ-point-selection failure; frontier is non-Polish / non-standard-Borel / irreducibly non-central. Sources in dependency order, each with the question to read it for.

1 Descriptive set theory

Kechris, *Classical Descriptive Set Theory* [6]. Borel / analytic / co-analytic hierarchy $\to$ Borel equivalence relations $\to$ the Glimm–Effros dichotomy; then uniformization.

- State Glimm–Effros precisely (non-smooth $\iff$ E_0 embeds $\iff$ no Borel two-valued transversal). Which word does not transport to clause (ii)? ("Borel.")
- Co-analytic uniformization-failure: why is its obstruction real-valued / over a standard-Borel base, hence Derr–Williamson-killed?

Gao, *Invariant Descriptive Set Theory* [3] (after Kechris). E_0 / E_∞ / hyperfinite / turbulence.

- Define hyperfinite; state Adams–Kechris. Why is E_∞ non-hyperfiniteness orthogonal to clause (ii)?

2 Large cardinals and forcing

Jech, *Set Theory* [4].

- Filters, ideals, ultrafilters: why is $\mathcal{F}_s = \{A : s(A) = 1\}$ not a filter on a non-Boolean concrete OML, even for a Dirac state? (MO_2: $a \cap b \neq \varnothing$ as sets yet $a \wedge b = 0$.)
- Measurable cardinals, Ulam matrices: re-derive the matrix by hand; locate the line needing non-orthogonal unions / comprehension.
- Forcing: why does set-forcing preserve measurability (Lévy–Solovay)?

Kanamori, *The Higher Infinite* [5] (after Jech, if the strength path calls). Measurable $\to$ ultrapower $\to$ elementary embedding $j\colon V \to M$.

- The UB route transports U along j, but U lives on Boolean $\mathcal{P}(\kappa)$; why is transport to orthogonal-only a costume?

3 Primary sources (read for technique)

- **Navara–Pták** [8]. Re-derive their ZFC non-Dirac σ-state. What makes $L_{f,g}$ support it, and why is it not a witness?
- **Derr–Williamson** [2] + **Maharam §8** [7]. Where is inner-regularity load-bearing in Thm D.6 / Maharam §8.2?
- **Blecher–Weaver** [1]. Identify the ultrafilter-from-masa step (Prop. 2.3); state the masa-free question it leaves open for the concrete two-valued case.

4 Notes

- **Companion plan (the positive-construction route):** the import sweep left one *lead* for building a witness — the realization theorems (Navara–Rogalewicz, Harding–Navara), pushed to concrete + σ-complete. That technique has its own dependency-ordered reading plan, `realization_technique_reading.tex`, anchored to the conjunction gap. This plan (DST + large cardinals) is for the impossibility / independence side; that one is for the positive side.
- Start: Kechris (Borel equivalence relations → Glimm–Effros).
- Definability-vs-existence first: bounds §3f and §3j, plus the Lean polarity gate.
- Operator algebra (masas, von Neumann algebras) is not on the critical path: the masa route is dead, so light functional analysis suffices for Blecher–Weaver.
- Chapter numbers are convenience pointers; navigate by topic.

References

[1] David P. Blecher and Nik Weaver. Quantum measurable cardinals. *Journal of Functional Analysis*, 273(5):1870–1890, 2017.

[2] Rabanus Derr and Robert C. Williamson. Systems of precision: coherent probabilities on pre-Dynkin-systems and coherent previsions on linear subspaces. *arXiv preprint arXiv:2302.03522*, 2023.

[3] Su Gao. *Invariant Descriptive Set Theory*, volume 293 of *Pure and Applied Mathematics*. CRC Press, Boca Raton, 2009.

[4] Thomas Jech. *Set Theory*. Springer, Berlin, 3rd millennium edition, 2003.

[5] Akihiro Kanamori. *The Higher Infinite: Large Cardinals in Set Theory from Their Beginnings*. Springer Monographs in Mathematics. Springer, Berlin, 2nd edition, 2009.

[6] Alexander S. Kechris. *Classical Descriptive Set Theory*, volume 156 of *Graduate Texts in Mathematics*. Springer, New York, 1995.

[7] Dorothy Maharam. Consistent extensions of linear functionals and of probability measures. In *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Vol. II: Probability Theory*, pages 127–147. University of California Press, Berkeley, 1972.

[8] Mirko Navara and Pavel Pták. Two-valued measures on σ-classes. *Časopis pro pěstování matematiky*, 108(3):225–229, 1983.